

New York Water Inspections – Dataflow Explanation

UNDERSTANDING THE PROCESS BEHIND WATER INSPECTION
DATA MANAGEMENT

OVERVIEW AND PURPOSE

Purpose of the New York Water Inspections Dataflow

Automated Data Ingestion

The dataflow automates ingestion from NYC Open Data, minimising manual handling and ensuring repeatable data refreshes.

Data Cleaning and Standardisation

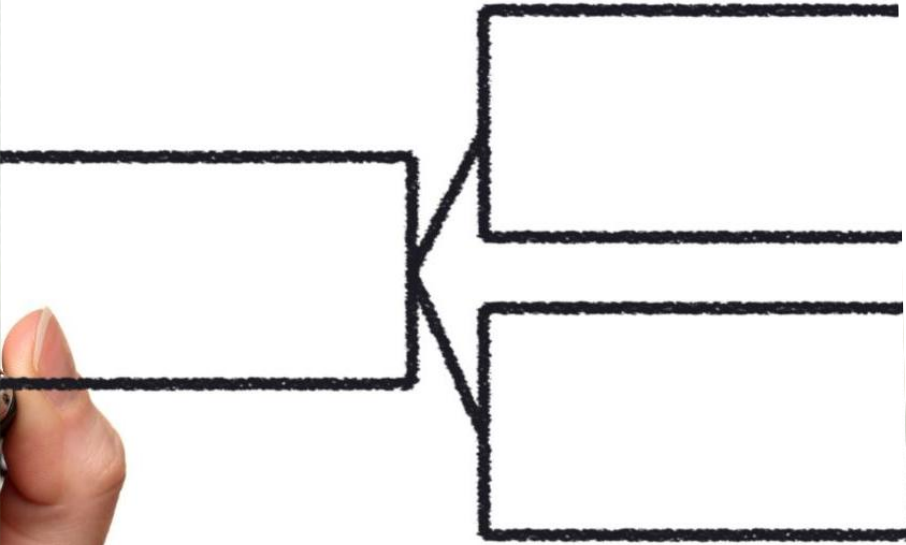
Systematic cleansing enforces data types and standardises dates and text for accurate time-based analysis.

Structured Output Design

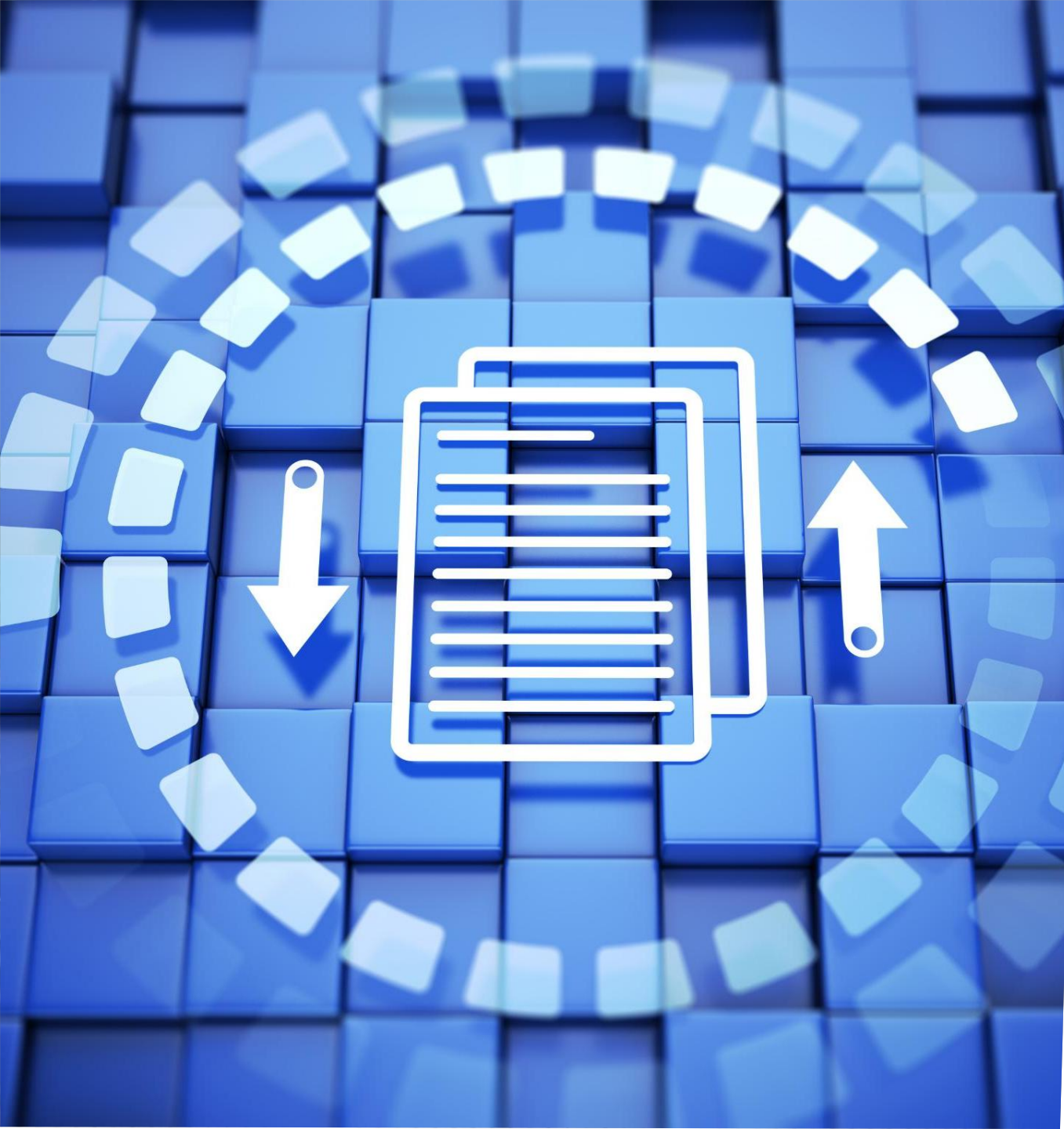
Separation into Buildings and Inspection tables supports star-schema design and reduces redundancy.

Data Enrichment for Consistency

Lookup tables enrich inspection records, improving readability and consistency for business users.



DATA SOURCES



Primary and Supporting Data Sources

Primary Data Source

The primary data source is a CSV feed from NYC Open Data containing detailed rooftop water tower inspection records accessed via HTTPS.

Supporting Lookup Tables

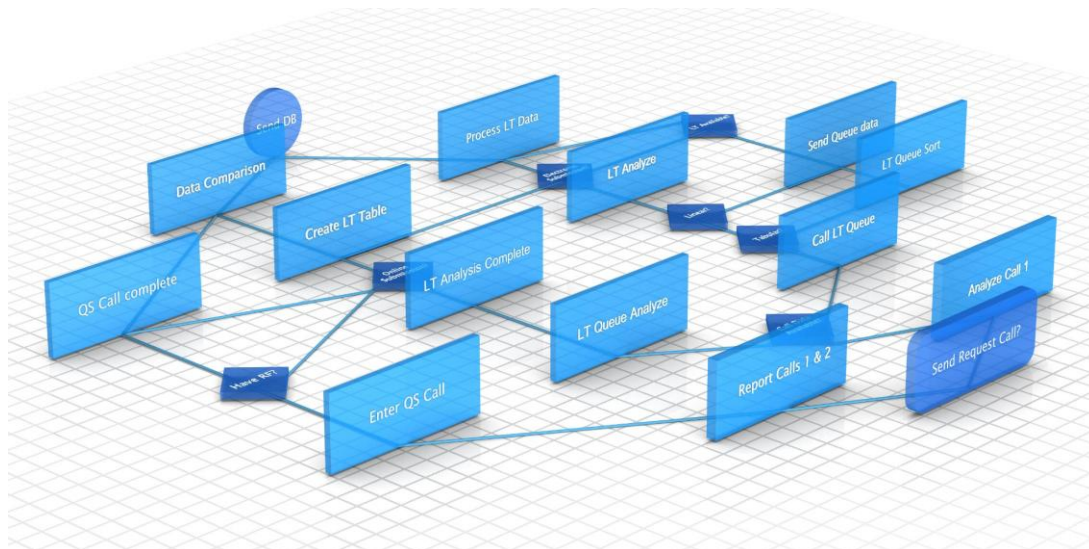
Supporting data comes from an Excel workbook on SharePoint with lookup tables mapping inspection firms and laboratories for consistency.

Data Governance and Integration

Combining authoritative public data with controlled internal references ensures traceability and consistent naming for reporting.

CORE TRANSFORMATIONS

Staging and Cleaning in the rows Query



Data Import and Structuring

The query imports CSV data and promotes the first row to headers, forming a usable table structure.

Applying Explicit Data Types

Explicit data types are assigned to fields for consistent behavior in numeric, geographic, and text data.

Parsing and Reformatting Dates

Text dates are split, reordered, and recombined into proper date values for accurate filtering and analysis.

Clean and Maintainable Output

Temporary columns are removed, producing a clean staged table for easier shaping and analysis downstream.

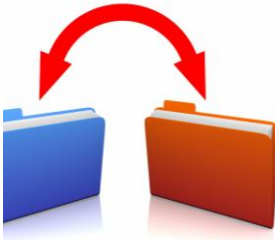
OUTPUT TABLES

Buildings Output Table



Static Building Attributes

The table contains static building data like BIN, address, borough, and coordinates, essential for modeling.



Deduplication Process

Duplicate records are removed based on BIN to ensure each building appears only once, improving data clarity.



Analytical Modeling Benefits

Isolating building attributes reduces data size and supports filtering, grouping, and mapping in semantic models.

Inspection Output Table

Focus on Inspection Details

The table retains essential inspection details by removing locational columns to highlight structural and biological findings.

Data Normalisation

Inspection firm and laboratory names are normalised to lowercase for consistent lookup and joining.

Duplicate Handling

A temporary index prevents record duplication, duplicates are removed before finalizing the clean inspection table.

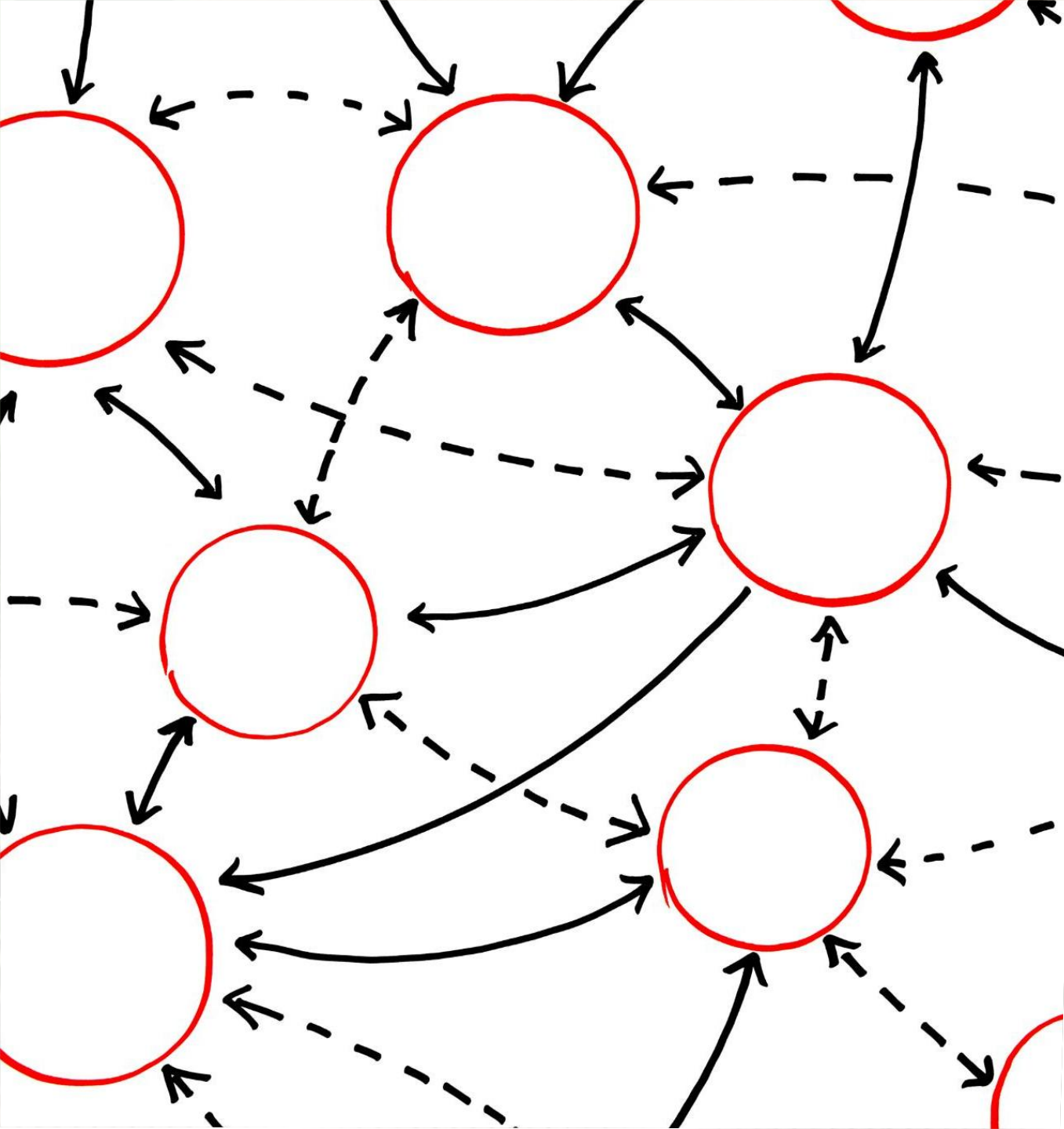
Clean Fact-Style Table

The final output is a clean, fact-style table enabling analysis by date, building, and compliance status.

The image shows a printed table with multiple columns and rows. The table is partially obscured by a red highlighter and a white ruler. The table contains numerical data, some of which is highlighted in red. The columns are labeled 'Date', 'In', 'Total', 'Hdcp.', and 'Net'. The rows are numbered 1 through 18. The table is a fact-style table, with columns representing different attributes and rows representing individual inspection records.

	Date	In	Total	Hdcp.	Net
377	385	4	8		
500	521	5	4		
422	459	4	6		
316	349	4	12		
182	230	3	16		
539	561	5	2		
160	201	3	18		
51	363	4	10		
8	199	3	14		
5	3268	35			
	6634	71			

ENRICHMENT AND FINAL MODEL



Lookup Enrichment and Resulting Model Shape

Lookup Enrichment Process

Lookup queries rename columns and normalize keys to map raw identifiers to readable names, improving data clarity.

Star Schema Model Shape

The final data model uses a star schema with Buildings as dimension and Inspection as fact table linked by BIN key.

Benefits for Power BI Models

This model enables intuitive filtering, efficient aggregation, and centralized transformation for reliable analytics.